Experiment 7 - Bragg's Law

Daire O'Brien - 24365676

November 4, 2026

1 Background

In 1912 German physicist Max von Laue discovered that when X-rays passed through a crystal sample diffraction and interferance occured. The effect was observed when a crystal was struck with a broad-band X-ray source and manifested as a ring-like pattern of dots on a photographic plate placed behind the crystal. As von Laue was investigating if X-rays were particles or waves, physicists William H. Bragg and William L.

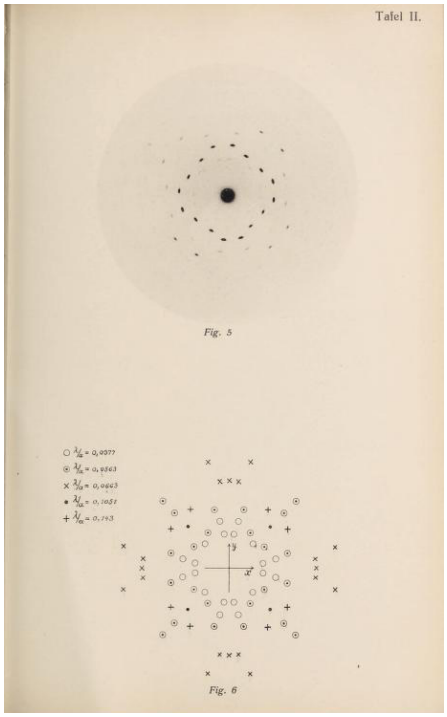


Figure 1: Interferance pattern from X-rays passing through a crystal sample [2]

Bragg investigated if the crystals could be used to reflect the X-rays.

Using an x-ray source, an adjustable apature and a turntable they were able to determine that scattering only occured when there was constructive interferance between the x-rays. They proposed that

$$n\lambda = 2d\sin\theta \tag{1}$$

was the condition for the reflected x-rays to be detected, where λ is the wavelength of the x-ray, d is the seperation between the atoms and θ is the angle of reflection.

More recently some of the assumption's that the original paper makes have been disputed as being ambiguous, particularily around the definition of the glancing angle. In 2025 physicist Pramode Bhattacharjee reformulated the relation to use the "unambiguous" angle of reflection instead:

$$n\lambda = -2d\cos\phi_1$$

where ϕ_1 is the unambiguous incidence angle.

2 References

1. William Henry Bragg, William Lawrence Bragg, 1913, *The reflection of X-rays by crystals*, Proc. R. Soc. A1 July 1913

2. W. Friedrich, P. Knipping, M. Laue, presented by A. Sommerfeld 1912, *Interference Phenomena for X-Rays, Theoretical section*, Proceedings of the Mathematical-Physical Class of the Royal Bavarian Academy of Sciences in Munich, Issue 2. English translation by J. J. Stezowski.

3. Pramode Ranjan Bhattacharjee, *Discovery of the lack of success of Bragg's law in respect of perfect quantification of the X-ray diffraction phenomenon,* Next Research, Volume 2, Issue 2, 2025, 100312, ISSN 3050-4759